

CASE 33

HEREDITARY PERIODIC FEVER SYNDROMES

Defects in intracellular receptors of innate immunity leading to an uncontrolled inflammatory response.

One of the first responses to infection is the generation of an inflammatory response by the innate immune system, which causes pain, redness, heat, and swelling (edema) in the infected tissues. Inflammation at the site of infection augments the killing of invading microorganisms by the front-line macrophages and causes changes in the walls of local blood vessels that enable additional effector molecules and cells, such as neutrophils, to enter the infected tissue from the blood (Fig. 33.1). Inflammation also induces local blood clotting, making a physical barrier to the spread of infection through the bloodstream. These changes are induced by a variety of inflammatory mediators that are released by macrophages and other cells of the innate immune system as a consequence of their recognition of pathogens.

Among these inflammatory mediators are pro-inflammatory cytokines that help to induce and maintain an inflammatory response. The main pro-inflammatory cytokines are interleukin-1 β (IL-1 β), tumor necrosis factor- α (TNF- α), and IL-6. In addition to provoking tissue inflammation, these cytokines also induce fever (Fig. 33.2). Fever is generally beneficial to host defense; most pathogens grow better at lower temperatures, and adaptive immune responses are more intense at elevated temperatures. IL-1 β and IL-6 also enhance the production of innate immunity proteins called the acute-phase reactants, which are synthesized by the liver and released into the blood.

Because unregulated inflammation is harmful to the host, the production of pro-inflammatory cytokines is tightly controlled, so they are only produced in significant amounts in response to infection. Cells involved in innate immunity, such as monocytes/macrophages, dendritic cells, and neutrophils, detect the presence of pathogens using both cell surface and intracellular receptors for microbial components.

TOPICS BEARING ON THIS CASE:

Inflammatory responses

Acute-phase responses

NOD-like receptors in innate immunity

Cytokine processing

The autoinflammatory syndromes

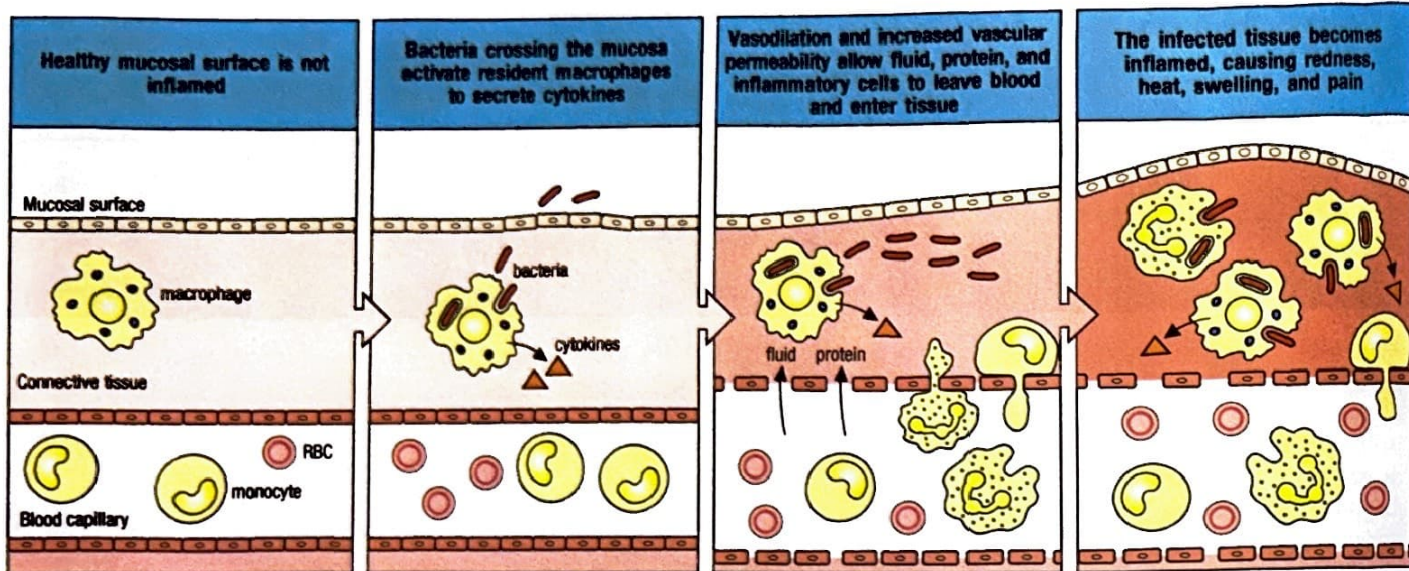
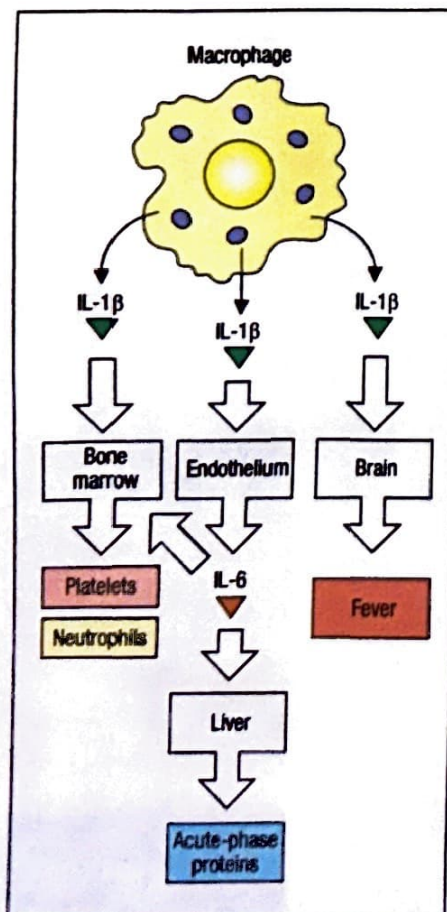


Fig. 33.1 Innate immune mechanisms establish a state of inflammation at sites of infection. Pathogens penetrating the skin or mucosal epithelium invade the underlying connective tissue. Once there, they activate resident effector cells such as macrophages to secrete chemokines, cytokines, and other

pro-inflammatory mediators, triggering the events leading to inflammation. Monocytes/macrophages and neutrophils are attracted into the inflamed tissues, where they contribute to enhancing and sustaining the inflammatory state.



Intracellular bacterial infections of macrophages are detected by a family of intracellular proteins called the NOD-like receptors (NLRs), which contain a nucleotide-binding oligomerization domain (NOD), and a group of leucine-rich repeats near the carboxy end. The NLRs include four subfamilies, distinguished by an additional protein domain found near the amino terminus: the NLRA (A for acid transactivation domain), NLRB (B for baculoviral inhibitory repeat), NLRP (P for pyrin domain), and NLRC (C for caspase recruitment domain; CARD) families. NLRs sense the cellular damage or stress that occurs with infection through the recognition of DAMPs (damage-associated molecular patterns) and PAMPs (pathogen-associated molecular patterns). Downstream signaling from NLRs leads to an enhanced inflammatory response and cell death.

Members of the NLRP subfamily are found in numerous cell types including monocytes/macrophages and neutrophils. Normally held inactive by accessory proteins in the cytoplasm, the NLRPs are activated by a variety of stimuli produced by stressed cells. For example, the toxins of bacteria such as *Staphylococcus aureus* and the intracellular pathogen *Listeria monocytogenes* cause an efflux of K^+ ions from affected cells. The resulting low levels of intracellular K^+ dissociate the inhibitory proteins from the NLRPs. One of the best characterized NLRPs is NLRP3, also known as cryopyrin because it was initially recognized as a pro-fever factor.

Fig. 33.2 Systemic effects of IL-1 β . Active IL-1 β is secreted by many cell types, including monocytes and macrophages. It enters the circulation and triggers IL-1 receptors on the hypothalamic vascular network, resulting in the synthesis of cyclooxygenase-2, which causes levels of prostaglandin E2 in the brain to rise, thus activating the thermoregulatory center for fever production. In the periphery, IL-1 β activates IL-1 receptors on endothelium, resulting in rashes and the production of IL-6. Circulating IL-6 stimulates liver cells to synthesize several acute-phase proteins, which cause an increased erythrocyte sedimentation rate. IL-1 β also acts on the bone marrow to increase the mobilization of granulocyte progenitors and mature neutrophils, resulting in peripheral neutrophilia. It also indirectly increases IL-6-induced platelet production, resulting in thrombocytosis, and causes a decreased response to erythropoietin, which results in anemia.

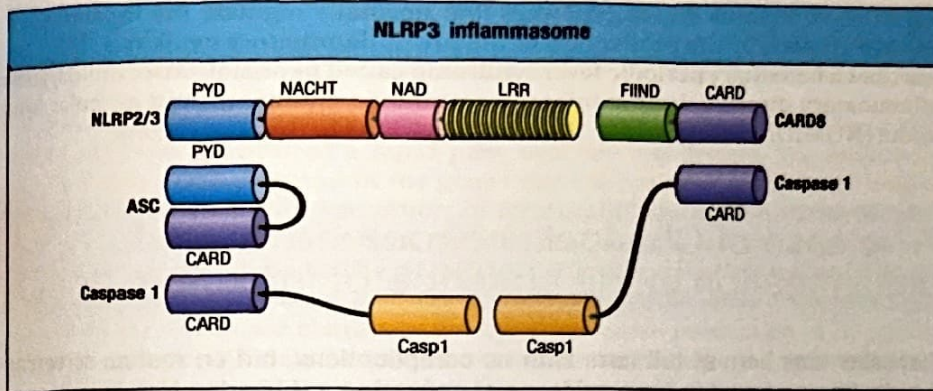


Fig. 33.3 A schematic of the NLRP3 inflammasome. NLRP3 has an amino-terminal pyrin domain (PYD), followed by a nucleotide-binding domain (NACHT), a NACHT-associated domain (NAD), and several leucine-rich repeats (LRRs). In the absence of stress or infection, the LRR domain is bound by inhibitory proteins that keep the NLRP inactive (not shown). Activated NLRP3 associates with the adaptor proteins ASC (apoptosis-associated speck-like protein containing a CARD) and CARD8, which have CARD domains that can recruit caspase 1 to the inflammasome.

In stressed cells, active NLRP3 forms part of a large protein complex named the inflammasome, for its role in inducing inflammation (Fig. 33.3). Assembly of the NLRP3 inflammasome leads to activation of protease caspase 1 and subsequent processing of IL-1 β and another pro-inflammatory cytokine, IL-18, into their active forms (Fig. 33.4).

A group of clinical syndromes, called the autoinflammatory syndromes, manifest as inflammation in the absence of any evidence of microbial infection. They are characterized by episodes of fever accompanied by systemic and localized inflammation, affecting the joints, skin, and the lining of the lungs, heart, and abdominal cavity. Unlike autoimmune conditions, the autoinflammatory syndromes do not involve self-reactive T cells or self-reactive antibodies (autoantibodies). As the name 'autoinflammatory' suggests, these syndromes are driven by unchecked inflammatory responses. In normal individuals, inflammatory cascades are tightly controlled, in large part by regulating the production of pro-inflammatory cytokines such as IL-1 β . In contrast, individuals affected by autoinflammatory syndromes develop adaptive immune responses to stimuli such as infection or vaccination, but they are unable to properly control inflammation once it is under way. In these affected individuals, inflammation may also occur in response to minor stimuli that would not elicit an inflammatory response in normal individuals.

The autoinflammatory syndromes include relatively common conditions such as systemic-onset juvenile rheumatoid arthritis (see Case 34), but most of the syndromes are rare monogenic diseases known as hereditary periodic fever syndromes.

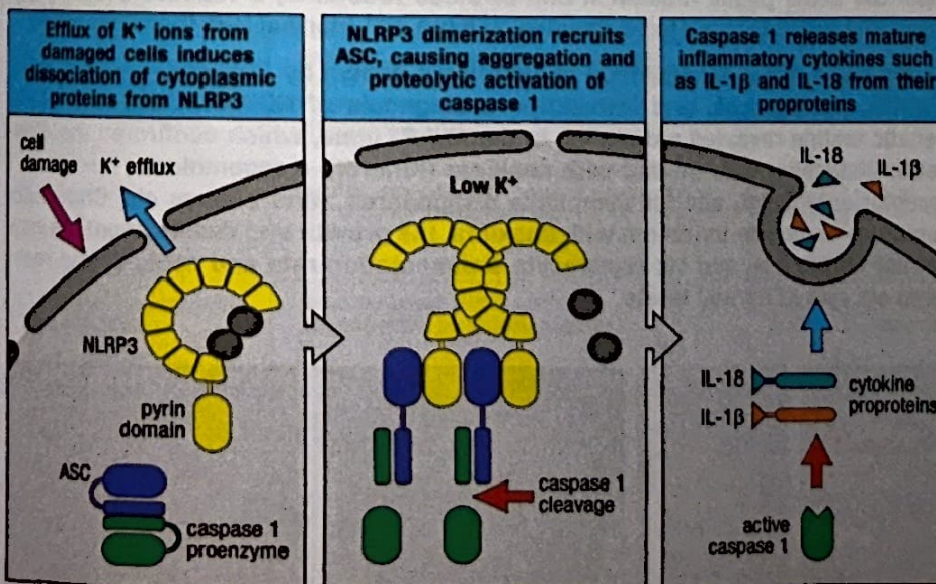


Fig. 33.4 NLRPs sense cellular damage and activate the processing of pro-inflammatory cytokines. Under normal physiologic conditions the LRR domain of NLRP3 associates with cytoplasmic proteins, which are presumed to prevent the dimerization of NLRP3 via the LRR region. When cells are injured or under stress, the efflux of K⁺ ions is thought to trigger the dissociation of these proteins from NLRP3, allowing its oligomerization (center panel). The pyrin domains of NLRP3 recruit complexes of the adaptor protein ASC, which associate with the caspase 1 proenzyme through their CARD domains. Aggregation of the proenzymes causes them to undergo autoactivation via proteolytic cleavage to form active caspases. Active caspase 1 cleaves the proprotein forms of pro-inflammatory cytokines to release the mature cytokines, which can then be secreted.

The mutated proteins impair pathways that normally regulate the inflammatory response, especially the production of the pro-inflammatory cytokines. This case describes a hereditary periodic fever syndrome called neonatal-onset multisystem inflammatory disease/chronic infantile neurologic cutaneous and articular syndrome (NOMID/CINCA), which is due to mutations in the *NLRP3* gene.

The case of Chancellor Krook: fever and inflammation in the absence of infection.

Chancellor was born at full term with no complications, but on routine screening at birth, he was found to have evidence of sensorineural hearing loss in both ears. At 1 week old, he developed intermittent fever accompanied by irritability, and a severe bacterial infection was suspected. His white blood cell count was elevated at $21,000 \mu\text{l}^{-1}$ (normal less than $10,000 \mu\text{l}^{-1}$). Cerebrospinal fluid (CSF) was sampled and also had an abnormally high white blood cell count of 500 ml^{-1} (normal for age is less than 20 ml^{-1}). Although no bacteria could be cultured from samples of blood, urine, or CSF, Chancellor was kept in hospital and treated with antimicrobial drugs because bacterial or viral meningitis was suspected. Despite this treatment, the fevers continued and no cause for it was identified. His mother noticed a raised, red, itchy rash all over Chancellor's body, which his physicians diagnosed as urticaria (hives), attributing it to a reaction to the antibiotics he had been given. The rash persisted, however, and Chancellor continued to have intermittent fevers. He was eventually sent home without a diagnosis.

The episodes of fever and irritability continued throughout Chancellor's first year, and his growth and development appeared unusually slow. Tests revealed progressive hearing loss, and he was given a hearing aid. In view of his continuing symptoms, Chancellor was referred to the rheumatology department at the Children's Hospital when he was 1 year old for testing for inflammatory diseases.

The examining physician noted Chancellor's prominent forehead and saddle-shaped nose (Fig. 33.5). His liver and spleen were enlarged, and examination of his joints revealed arthritis in his knees. A diffuse, red, urticaria-like rash covered his trunk, back, legs, and arms. Laboratory tests showed an elevated erythrocyte sedimentation rate of 35 mm h^{-1} (normal $0\text{--}20 \text{ mm h}^{-1}$). Sedimentation of red blood cells is hastened when the concentration of certain blood proteins is increased, among them the acute-phase reactants. One of these reactants, C-reactive protein, was measured and was found to be elevated, at 8 mg dl^{-1} (normal less than 0.50 mg dl^{-1}).

On the basis of the constellation of symptoms shown by Chancellor—fever, rash, arthritis, hearing loss, and irritability—a diagnosis of NOMID/CINCA was made. Genetic testing revealed a mutation in the *NLRP3* gene, which confirmed the diagnosis. Chancellor was treated with anakinra (Kineret), a recombinant human IL-1 receptor antagonist, and his symptoms disappeared. Now 2 years old, Chancellor continues to require treatment with anakinra. His growth and development are now normal for his age, and his erythrocyte sedimentation rate and acute-phase reactants are also at normal levels.

Infant with recurrent fever and irritability. No infection detectable.

Hearing loss, arthritis, recurrent fever. Autoinflammatory disease?

Mutation in *NLRP3* gene.

Fig. 33.5 Clinical features of NOMID/CINCA. Panel a, dysmorphic facial features: frontal bossing, saddle nose, and long philtrum of upper lip; panel b, urticarial rash; panel c, enlarged liver and spleen with distended abdomen (markings show the lower margins of the spleen (right) and the liver (left)); panel d, fundoscopic image of the optic disk showing papilledema.



Hereditary periodic fever syndromes.

The discovery of distinct disorders classified as hereditary periodic fever syndromes has been growing at a rapid pace over the last decade. The syndromes (Fig. 33.6) differ clinically and in the genes that are mutated, but they are unified by the involvement of both systemic and localized inflammation. Although classified as 'periodic' fever syndromes, the recurrent bouts of inflammation often do not occur at regular intervals. The NOMID/CINCA diagnosed in Chancellor is one of three related but clinically distinct 'cryopyrinopathies' among the hereditary periodic fevers, which are characterized by an excessive production of IL-1 β . The effects of excess IL-1 β include fever, rash, excess neutrophils in the blood (neutrophilia), high levels of blood platelets (thrombocytosis), and the production of acute-phase reactants, all of which were seen in Chancellor's case. The other cryopyrinopathies are Muckle-Wells syndrome and familial cold autoinflammatory syndrome (FCAS), and all three disorders result from mutations in the cryopyrin-encoding gene *NLRP3*. The exact mechanism by which the mutations found in patients with cryopyrinopathies promote inflammation is still under investigation.

Disease	Clinical features	Functional defects	Inheritance	Defective gene (protein)
Familial Mediterranean fever (FMF)	Recurrent fever, serositis (inflammation of the pleural and/or peritoneal cavity) responsive to colchicine, amyloidosis. Predisposes to vasculitis and inflammatory bowel disease	Decreased production of pyrin leading to increased IL-1 β processing and secretion; decreased macrophage apoptosis	Autosomal recessive	<i>MEFV</i> (pyrin)
TNF-receptor associated periodic syndrome (TRAPS)	Recurrent fever, serositis, rash, and ocular or joint inflammation	Reduced levels of soluble TNF receptor leading to excess TNF- α in the blood	Autosomal dominant	<i>TNFRSF1A</i> (TNF 55 kDa receptor/TNFR-I)
Hyper-IgD syndrome (HIDS)	Periodic fever and leukocytosis with high levels of IgD in serum	Mevalonate kinase deficiency affecting cholesterol synthesis; pathogenesis of disease unclear	Autosomal recessive	<i>MVK</i> (mevalonate kinase)
Pyogenic arthritis, pyoderma gangrenosum and acne (PAPA)	Periodic fever, myalgia (muscle pain), rash, acute-phase response	Sequestration of pyrin by CD2BP1 leading to increased IL-1 β processing and secretion	Autosomal dominant	<i>PSTPIP</i> (CD2-binding protein 1/CD2BP1)
Muckle-Wells syndrome (MWS)*	Urticaria, hearing loss	Constitutive activation of cryopyrin leading to increased caspase 1 activation and IL-1 β processing and secretion	Autosomal dominant	<i>NLRP3</i> (cryopyrin)
Familial cold autoinflammatory syndrome (FCAS)*	Non-pruritic urticaria, arthritis, chills, fever, and leukocytosis after exposure to cold			
Neonatal onset multisystem inflammatory disease/chronic infantile neurologic cutaneous and articular syndrome (NOMID/CINCA)*	Neonatal-onset rash, chronic meningitis, hearing loss, arthropathy with fever and inflammation responsive to the IL-1R antagonist anakinra			
Blau syndrome	Uveitis, synovitis, rash, cranial neuropathy. It is frequently associated with Crohn's disease	Mutations of <i>NOD</i> , possibly disrupting interaction with bacterial cell-wall muramyl peptides and NF- κ B signaling	Autosomal dominant	<i>NOD2</i> (NOD2)
Deficiency of IL-1 receptor antagonist (DIRA)	Neonatal-onset sterile multifocal osteomyelitis, periostitis, pustulosis	Mutations of IL-1R antagonist, leading to unopposed action of IL-1 β	Autosomal recessive	<i>IL1RN</i> (IL-1 receptor antagonist)
Chronic recurrent multifocal osteomyelitis and congenital dyserythropoietic anemia	Multifocal chronic osteomyelitis, anemia, cutaneous inflammatory disorders	Undefined	Autosomal recessive	<i>LPIN2</i>

Fig. 33.6 The spectrum of hereditary autoinflammatory syndromes. The cells primarily affected in these diseases are granulocytes, especially neutrophils, and monocytes and macrophages. Chondrocytes are also affected in NOMID/CINCA.

*The cryopyrinopathies. All three syndromes are associated with similar mutations in the *NLRP3* gene; the disease phenotype in any individual depends on the modifying effects of other genes and environmental factors.

The mutant NLRP3 proteins have been found to be constitutively active and hyperresponsive to microbial antigens or to physical stimuli such as cold, resulting in exaggerated secretion of IL-1 β and IL-18.

NOMID/CINCA patients typically have intermittent fevers associated with persistent urticaria, arthritis-like symptoms, and central nervous system inflammation that ranges from papilledema (swelling of the optic disk, where the sensory fibers from the retina exit from the eye), increased numbers of inflammatory cells in the CSF (causing a chronic meningitis), to sensorineural hearing loss (deafness due to a defect in nerve conduction or perception by the brain of the auditory impulse).

Familial Mediterranean fever (FMF) is the most common hereditary periodic fever syndrome and is associated with mutations in the gene *MEFV*, which encodes a protein called pyrin, named after the fever characteristic of the disease. Individuals with FMF have frequent fevers, abdominal pain, chest pain, rashes, and arthritis. Mutations in the *PSTPIP1* gene, which encodes PSTPIP1, a pyrin-interacting protein, are associated with the syndrome of pyogenic arthritis, pyoderma gangrenosum, and acne (PAPA). These mutations accentuate the interaction between pyrin and PSTPIP1, and it has been suggested that the sequestration of pyrin results in excess secretion of IL-1 β .

TNF receptor-associated periodic syndrome (TRAPS, originally known as familial Hibernian fever) presents with longer episodes of fever than FMF, abdominal pain, and a rash. This disorder is due to mutations in the gene encoding the TNF receptor 55 kDa subunit (*TNFRSF1A*) that lead to excess TNF- α in the circulation.

Mevalonate kinase deficiency (MKD), also known as hyperimmunoglobulinemia D with periodic fever syndrome (HIDS), was originally differentiated from FMF by the detection of elevated levels of IgD. Attacks of MKD tend to last 7 days and recur every 4–8 weeks. A more serious condition, mevalonate aciduria, caused by severely impaired mevalonate kinase activity, causes persistent fevers, chronic inflammation, poor growth, and severely impaired neurologic development in children.

Other autoinflammatory syndromes include Blau syndrome, which presents with uveitis, synovitis, skin rash, and cranial neuropathies, and is associated with Crohn's disease. It is due to mutations in the *NOD2* gene that may interfere with the binding of the NOD2 protein, a member of the NLRC family, to bacterial cell-wall components and subsequent signaling through NF κ B.

One of the most serious long-term complications of the periodic fever syndromes is the development of systemic amyloidosis, which occurs when misfolded fragments of serum amyloid, an acute-phase reactant, are deposited in tissues. The kidneys, gastrointestinal tract, adrenal glands, spleen, testes, and lung are most often affected, but the liver, heart, and thyroid gland can also be involved. Amyloidosis occurs most commonly in patients with FMF, but it can also affect patients with TRAPS and Cryopyrinopathies. Treatment with the drug colchicine, and to a lesser extent with TNF- α inhibitors and the IL-1 receptor antagonist anakinra, has been shown to prevent the development of amyloidosis.

Questions.

- 1 Why has the drug anakinra proved a successful treatment for some of the autoinflammatory syndromes?
- 2 What is the proposed mechanism of action for colchicine in the treatment of the autoinflammatory syndromes?
- 3 What cytokine, other than IL-1 and IL-1 β , would you expect to be elevated in the setting of caspase 1 activation?